Abtin Aryanpour

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Education

University of Waterloo — Waterloo, ON
Candidate for BASc in Computer Engineering, Co-op Program — *Sep. 2026 – Apr. 2031*

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS
Frameworks: PyTorch, Django, Streamlit, Tailwind CSS
Libraries: NumPy, OpenCV, scikit-learn, shapely, jQuery
Embedded: Raspberry Pi Pico SDK, Arm Cortex-M33, PIO, DMA, I2S, PWM
Tools & Platforms: Git, GitHub Actions (CI), pytest, uv, CMake, SQLite, AWS (EC2, CloudFront)

Projects

Andon: Machine-Sound Anomaly Detection | *Python, C, Raspberry Pi Pico 2* — Oct. 2026 – Present

- Built a system that detects failing machines by sound, training a PyTorch CNN only on healthy audio; it reached 92.13% mean AUC on held-out DCASE 2020 data vs. 79.70% for the DCASE autoencoder baseline (3 seeds)
- Validated the evaluation by reproducing the DCASE baseline within 0.88 AUC points on all 6 machine types
- Developed a streaming green/yellow/red alert using control-chart (SPC) limits, catching 65.57% of 30 s faults with a 10.57 s mean delay at 5.53 false alarms/hour on held-out clips
- Quantized the CNN to int8 for a Cortex-M33 (98.7 KB from 383.0 KB, at 92.08% AUC) and ported the log-mel front end and int8 inference to C, running on a Pico 2 with 0 of 24,576 int8 inputs differing from Python
- Built a Pico 2 rig with a MOSFET fan kill switch (40/40 trials) and I2S mic; recording a 3-fan dataset (1 of 3 sessions done)

CyberShield: Full-Stack Discussion Forum | *Python, Django, JavaScript, Tailwind CSS* — Mar. 2025

- Built a full-stack forum as the only developer on a 3-person team using Django, jQuery (AJAX) and SQLite, winning 1st of 32 teams at CNLC 2025
- Implemented voting, sorting, filtering and fuzzy search (Fuse.js) through JSON endpoints, with no page reloads
- Secured accounts with Django auth, CSRF protection, reCAPTCHA and moderator roles; deployed on AWS

Experience

Autonomy Software Member — Sep. 2026 – Present
Waterloo Aerial Robotics Group — *Waterloo, ON*

Groundside Image Stitcher — PR #215 (in review)

- Built it as sole developer in Python (PyTorch, OpenCV, XFeat, RANSAC), merging out-of-order, GPS-tagged photos from 3 simulated drones into one georeferenced map; 112 tests
- Fused image matches with GPS and drone yaw, aligning photos to 4.3–8.3 cm on average at 3 m of simulated GPS error
- Cut processing from 0.42–0.75 s to 0.12–0.19 s per photo by using drone yaw to replace four-way feature extraction, in simulation
- Designed a bundle-adjustment-style least-squares solver, cutting simulated misalignment from 13.6–69.8 to 4.0–8.3 cm
- Hardened matching against 180-degree turns and repetitive terrain, forming one map on 10/10 simulated runs
- Built lens undistortion and blur, exposure and tilt checks through 2 review rounds; the blur check sorted 2,730 simulated photos under 9 noise and blur conditions with 0 errors

Swarm Spacing (Lloyd's Algorithm) — PR #247 (in review)

- Built a module that spreads drones evenly over any GPS-bounded region using Voronoi cells, handling concave areas and stray drones; reduced spacing unevenness from 0.46 to 0.06 in simulation

Homework Helper & Assistant Teacher — Sep. 2022 – Jul. 2025
Spirit of Math Schools Inc. — *Richmond Hill, ON*

- Ran a drop-in homework classroom as one of two staff (2024–2025), helping about 30 students per session
- Taught 1–2 classes a week of up to 16 students (2022–2024), covering lessons and contest-style algebra